1. A machine learning model that learns from the observations we make and makes predictions about the values of the objective functions on unseen data points
2. An optimization policy that makes decisions about where to make the next observation by evaluating the objective in order to locate the optimum

We introduce each of these two components in this section.

1.2.1 Modeling with a Gaussian process

Bayesian optimization works by first fitting a predictive machine learning model on the objective function we are trying to optimization—sometimes this is called the surrogate model, as it acts as a surrogate between what we believe the function to be from our observations and the function itself. The role of this predictive model is very important, as its predictions inform the decisions made by a Bayesian optimization algorithm and therefore directly affect optimization performance.

In almost all cases, a Gaussian process (GP) is employed for this role, which we will examine in this subsection. To an ML practitioner, GPs might not be the most popular class of models, compared to, say, ridge regression, decision trees, support-vector machines, or neural networks. However, as we will see time and again throughout this book, GPs come with a unique and essential feature: they do not produce point-estimate predictions like the models listed above; instead, their predictions are in the form of *probability distributions*.

Probabilistic predictions are key in Bayesian optimization, allowing us to quantify uncertainty in our predictions, which in turn helps trade off risk and reward when making decisions.

Let's first see what a GP looks like when we train it on a data set. As an example, say we are interested in training a model to learn from the data set in Table 1.3, which is visualized as black x's in Figure 1.4.

Table 1.3 Example regression data set corresponding to Figure 1.4.

Training data point	Label
1.1470	1.8423
-4.0712	0.7354
0.9627	0.9627
1.2471	1.9859

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